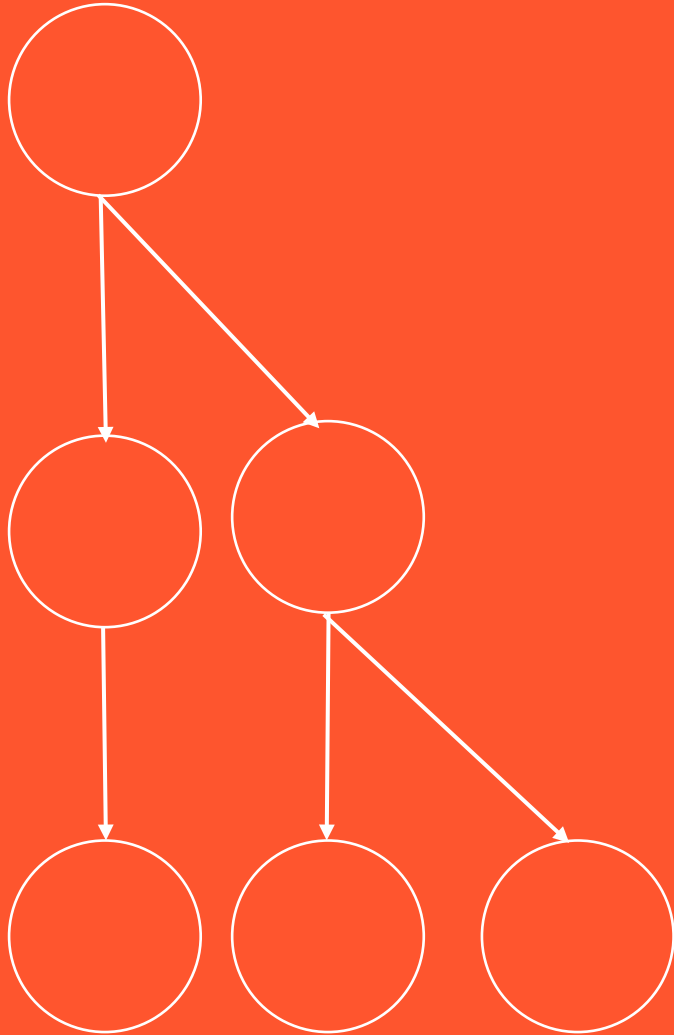


BCOG 100

Neural Circuits

Lecture 2: Adaptation and Plasticity



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 5-2

Password: hebb

Question: tell me something about hebbian learning



Adaptation and Plasticity

Recap & Overview

Last time: Neural Circuits = building blocks of brain function

Today: circuits aren't fixed — they adapt & learn



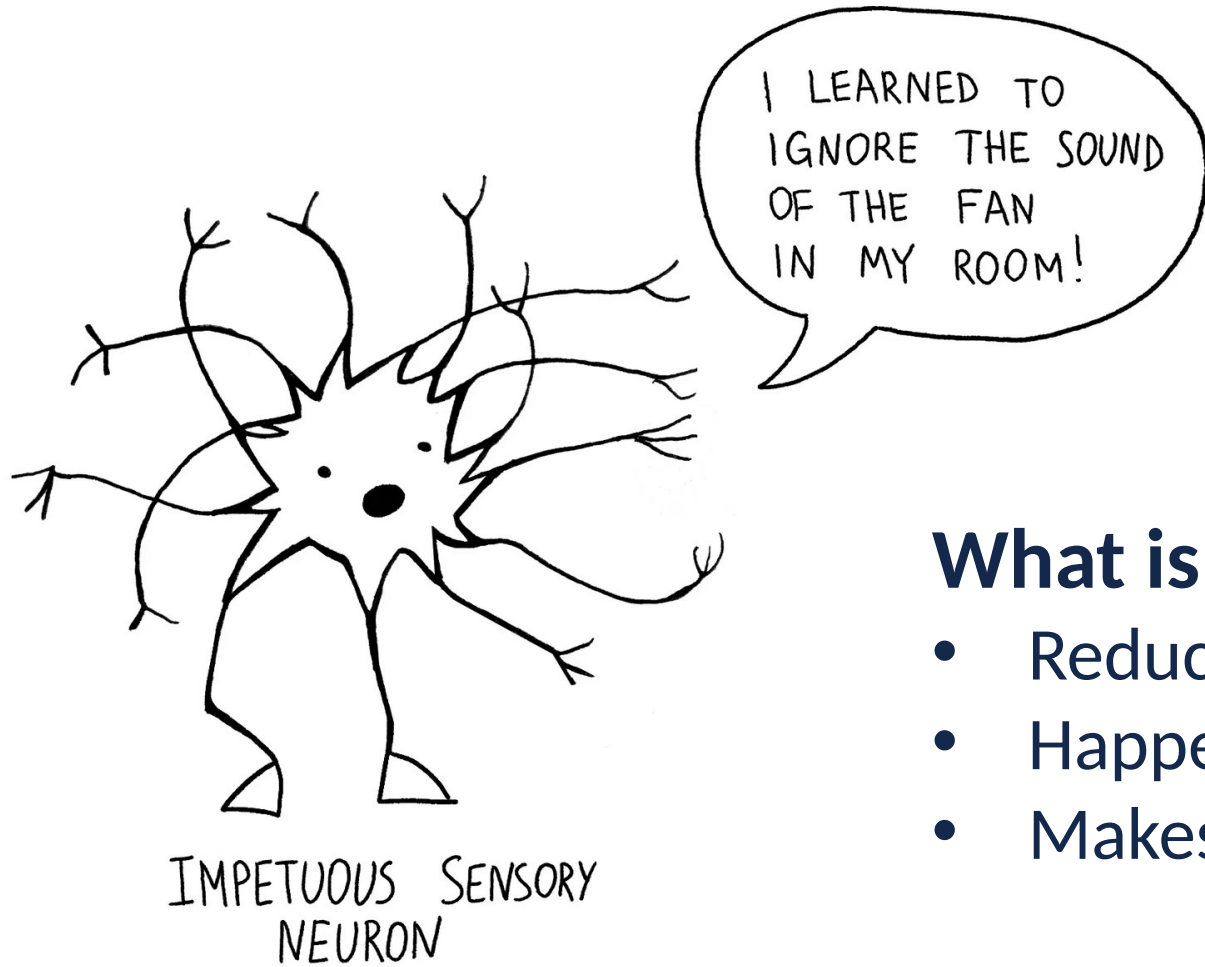
Adaptation and Plasticity



Why Hard-Wired Circuits Aren't Enough

- Fixed wiring = same response every time
- Organisms need flexibility
- Example: ignoring harmless, repeated touch

Neural Adaptation



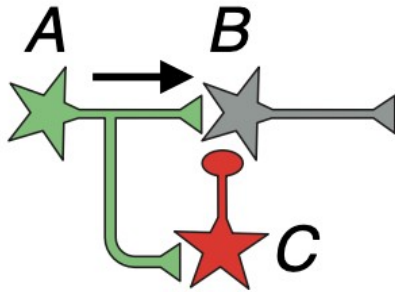
What is Neural Adaptation?

- Reduced response to repeated stimulus
- Happens quickly, often reversible
- Makes perception & behavior more efficient

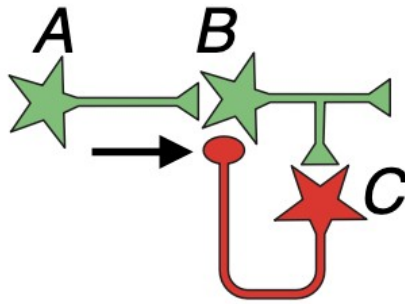
Neural Adaptation

B

feedforward
inhibition



feedback
inhibition



How Does Adaptation Work?

- **Single neuron:** reduced firing over time
- **Circuits:** inhibitory feedback dampens response
- Both mechanisms result in less response to repeated input

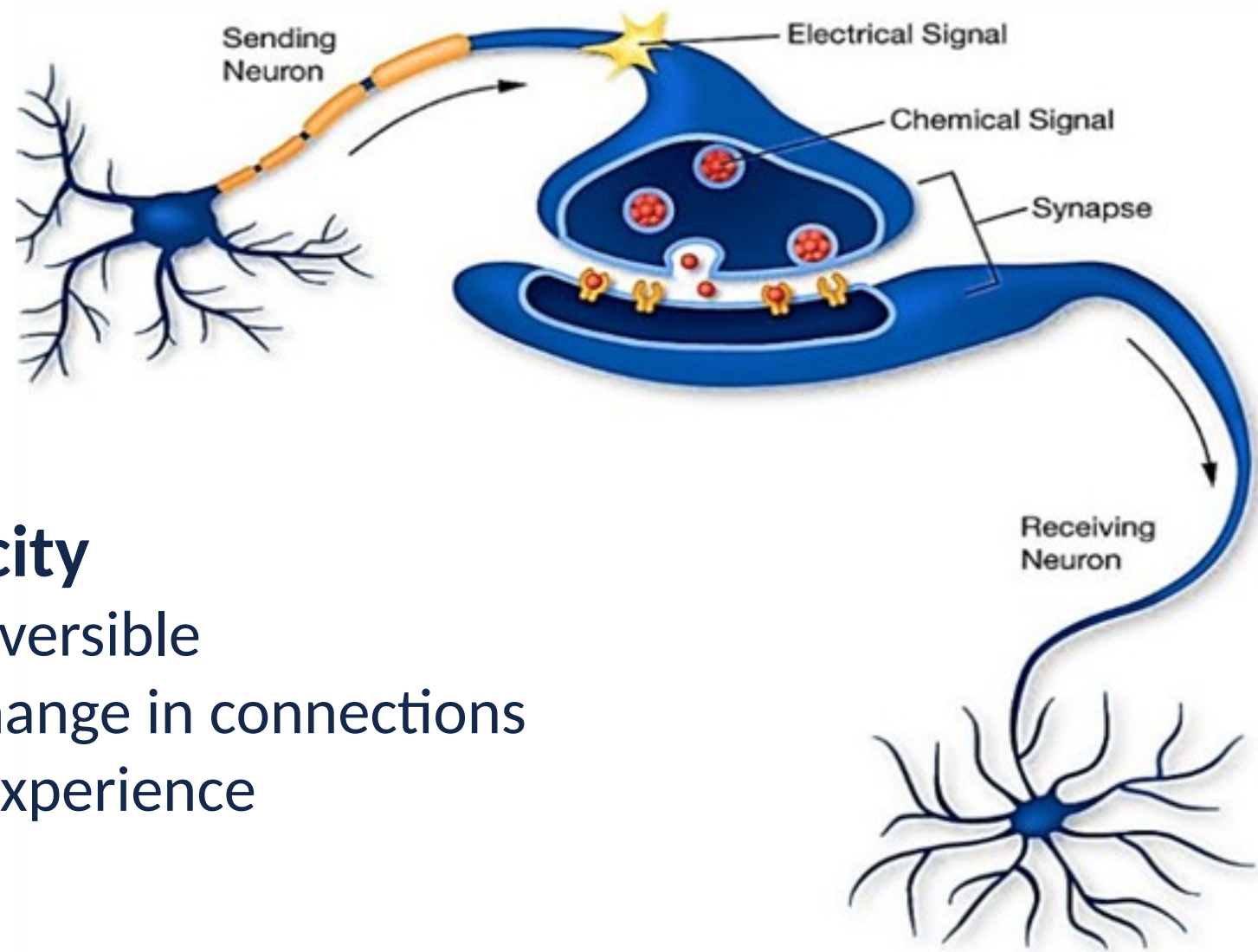
Neural Adaptation



Human Examples of Adaptation

- Smell: perfume fades after minutes
- Touch: clothes felt at first, then ignored
- Vision: afterimages & light/dark adjustment

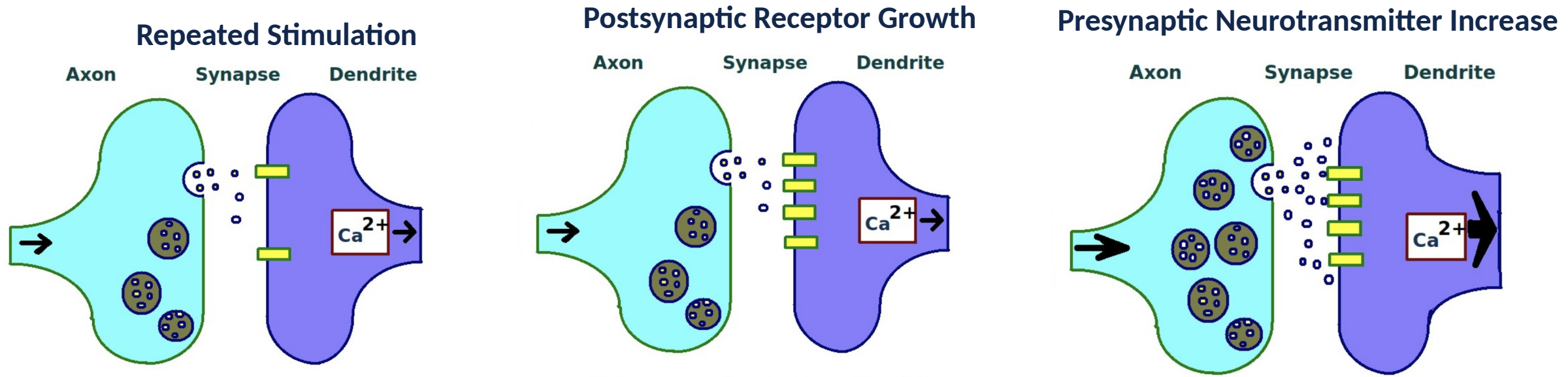
Learning and Plasticity



From Adaptation to Plasticity

- Adaptation = short-term, reversible
- Plasticity = longer-lasting change in connections
- Circuits can be rewired by experience

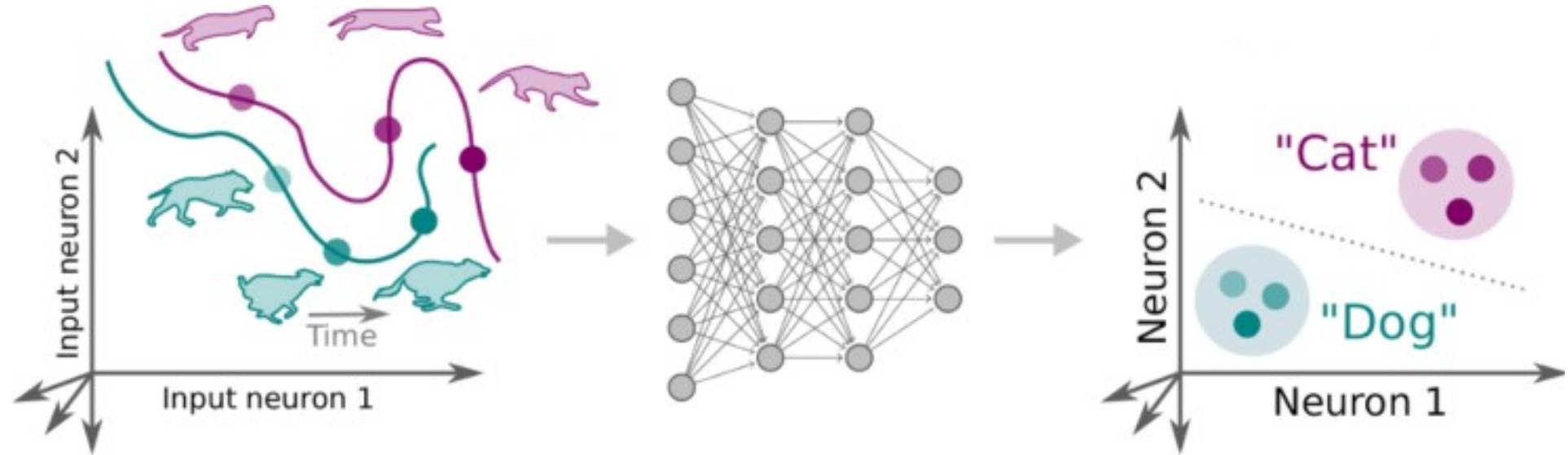
Learning and Plasticity



Synaptic Plasticity Basics

- **Long term potentiation (LTP):** repeated activity strengthens synapse
- **Long term depression (LDP):** uncoordinated activity weakens synapse

Learning and Plasticity



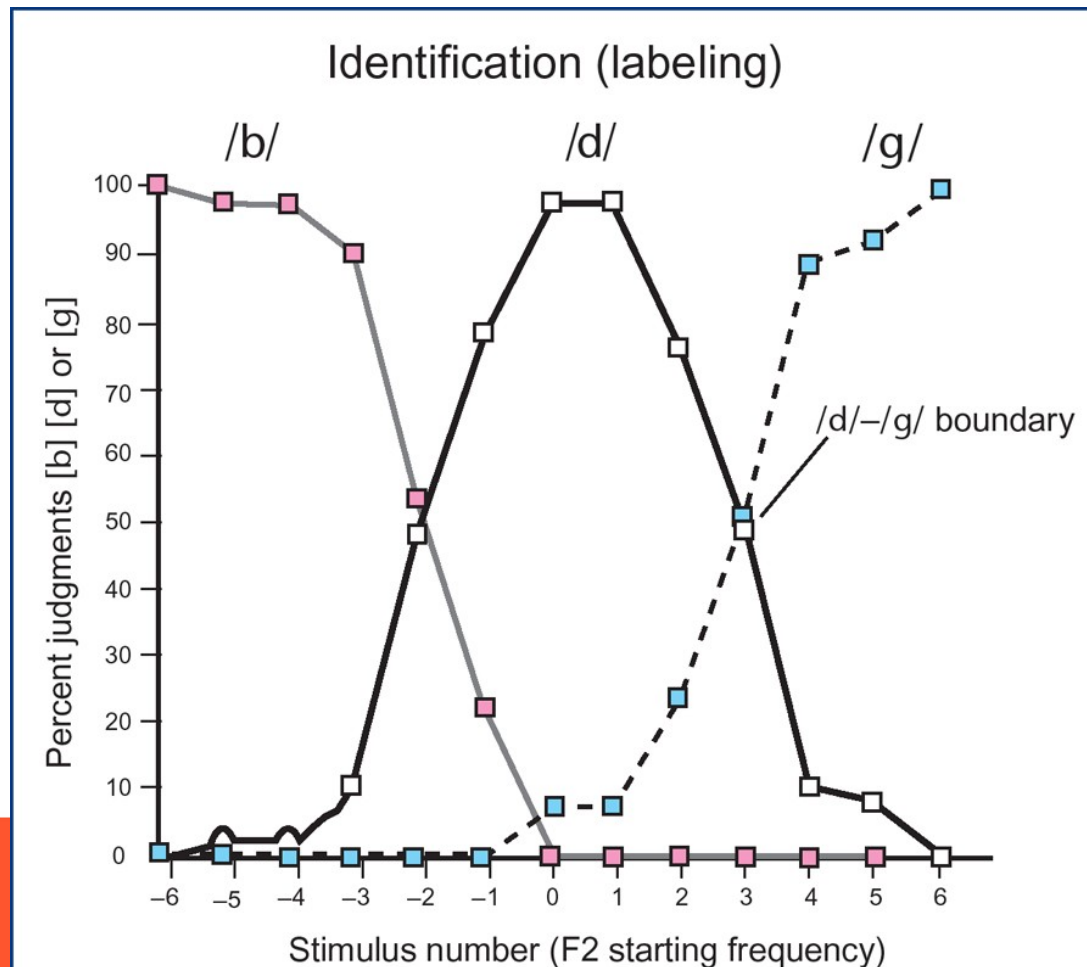
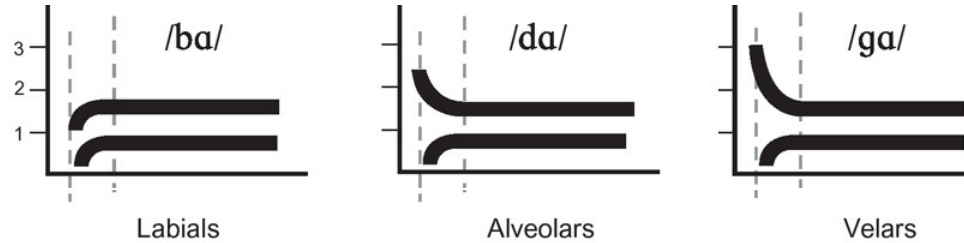
Hebb's Rule

- “Neurons that fire together, wire together”
- Coincidence strengthens connections
- Learning = capturing regularities in experience

Formula

- Δw_{ij} = the change in the weight from presynaptic neuron i to postsynaptic neuron j
- η = the learning rate (a small constant that controls how fast learning happens)
- x_i = activity of the presynaptic neuron (input)
- y_j = activity of the postsynaptic neuron (output)

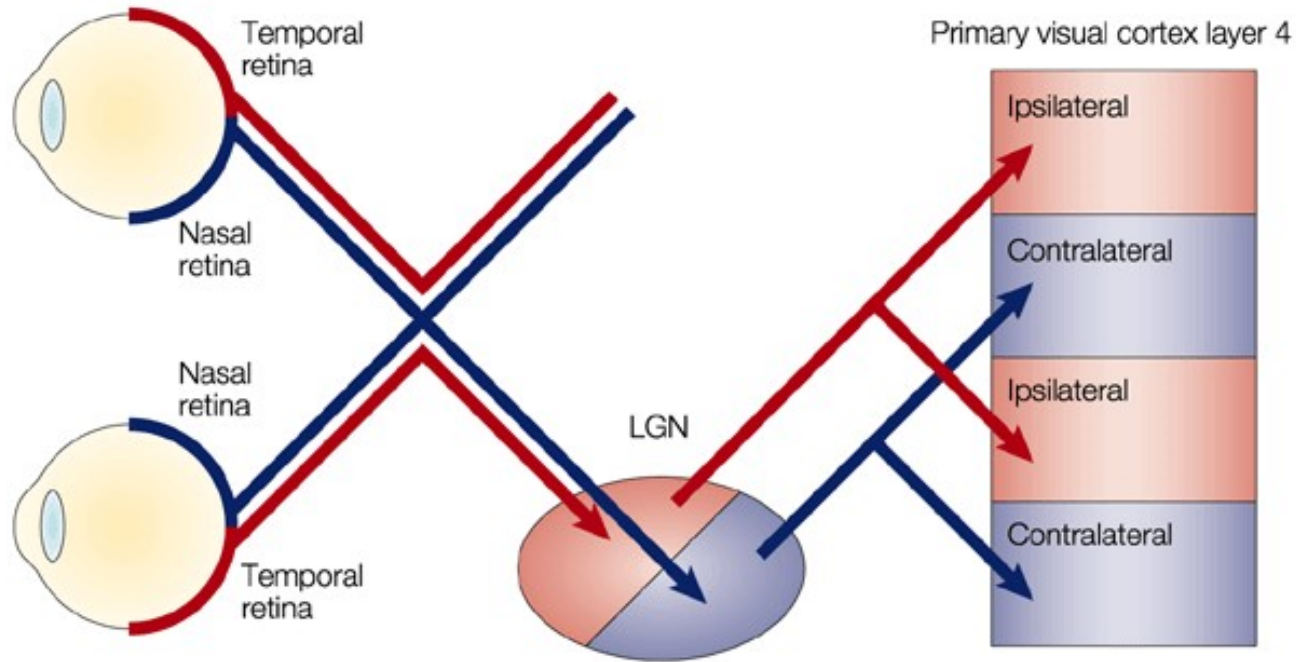
Learning and Plasticity



Hebb's Rule in Action

- *C. elegans*: touch → stronger withdrawal reflex
- Humans: speech sounds → tuned categories
- Circuits shape what gets processed

Learning and Plasticity



Nature Reviews | Neuroscience

Developmental Examples

- Critical periods: experience shapes circuits
- Language: phoneme tuning in infancy
- Vision: ocular dominance columns

Learning and Plasticity



Limits of Hebbian Learning

- Strengthens co-occurrence, but can't unlearn
- Perseveration: stuck responses
- Example: AND vs. OR situations

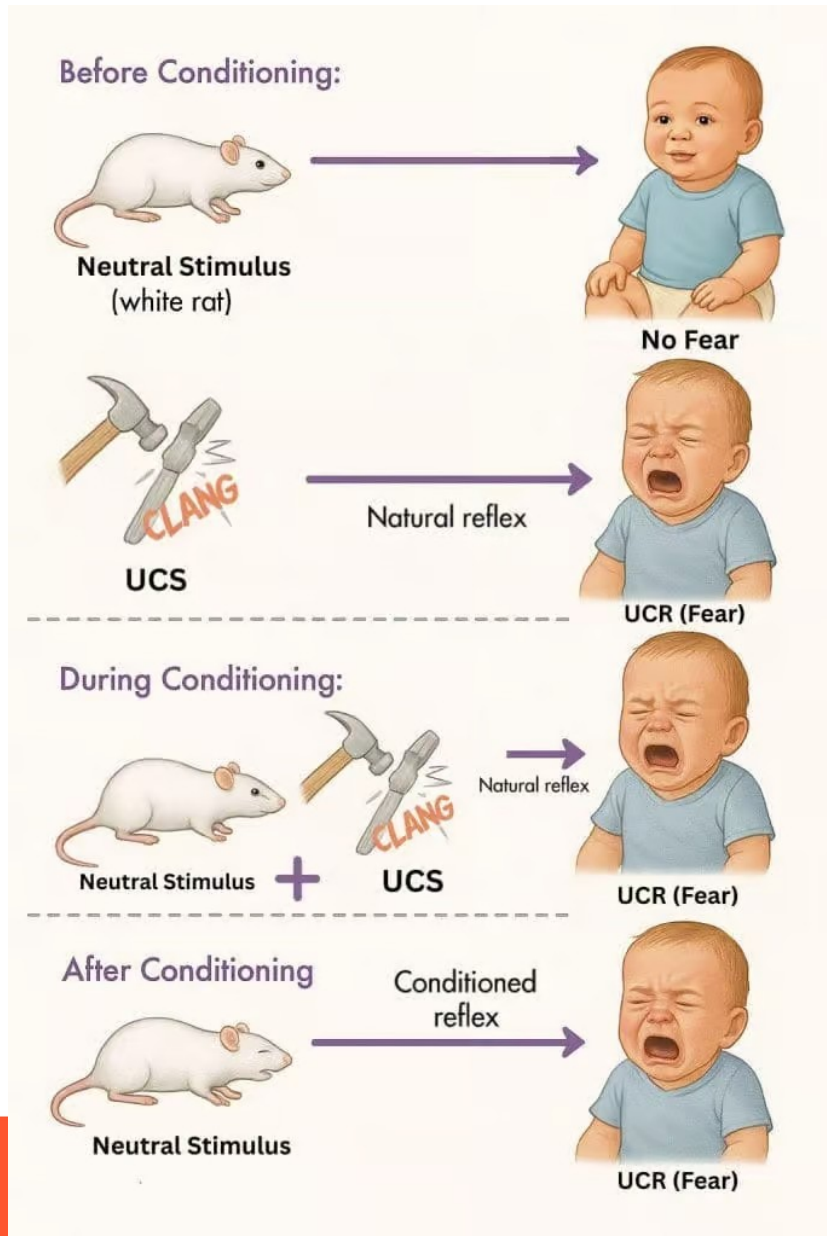
Learning and Plasticity



Beyond Simple Hebbian Learning: Supplements

- Normalization rules (e.g., Oja): balance weights: “fire together → stronger, but only if it’s a reliable co-activation. Otherwise, the weight shrinks back.”
- Error-driven learning: weakens wrong predictions
- Reinforcement signals: strengthen what’s rewarded

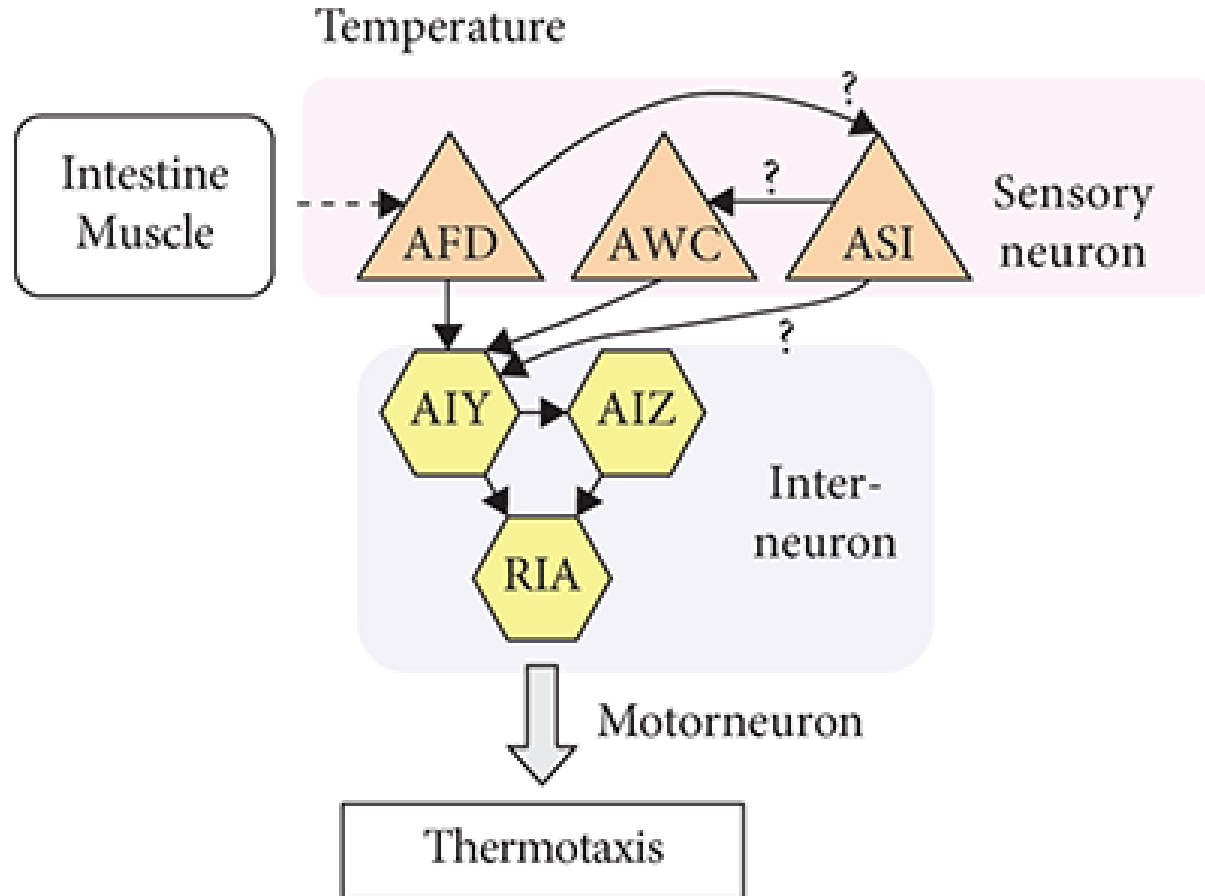
Learning and Plasticity



Hebbian Learning and Classical Conditioning

- Neutral cue + important event → learned link
- Pavlov's dogs: sound + food → salivation
- Hebbian principle: co-activation strengthens circuit

Learning and Plasticity



Conditioning in *C. elegans*

- Worms can form simple associations
- Odor + food → attraction
- Touch + stimulus → altered response

Learning and Plasticity



Classical Conditioning in Humans

- Conditioning underlies complex human learning
- Emotional responses: phobias, stress
- Social & cultural cues: advertising, habits

Adaptation and Plasticity Summary

Wrap-Up & What's Next

Today:

- Circuits aren't fixed — they adapt & learn
- Adaptation: short-term, reversible
- Hebbian learning: long-term, builds associations

Next: circuits as logic gates & computation

